Nils Sturma

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`https://nilssturma.github.io/`

Academic Employment

09/2025 - today	**Postdoctoral Researcher at EPFL** Chair of Biostatistics (Mats Stensrud) Also collaborating with the Chair of Mathematical Statistics (Victor Panaretos)
10/2024 - 08/2025	**Postdoctoral Researcher at Technical University of Munich** Research group Mathematical Statistics (Mathias Drton)

Education

04/2021 - 09/2024	**Ph.D. in Mathematical Statistics** Technical University of Munich (TUM), Germany Advisor: Mathias Drton Thesis: Identifiability and Statistical Inference in Latent Variable Modeling Grade: summa cum laude; date of defense: September 24, 2024.
09/2022 - 12/2022	**Research stay at MIT/ Broad Institute in Cambridge, USA** Working with Caroline Uhler Project: Unpaired Multi-Domain Causal Representation Learning
10/2018 - 01/2021	**Master in Mathematical Finance and Actuarial Science** Technical University of Munich (TUM), Germany Thesis: Testing Algebraic Constraints on Statistical Parameters Final grade: 1.2, with distinction (scale: 1 best, 6 worst)
02/2020 - 07/2020	**Semester abroad at University of Melbourne, Australia**
10/2014 - 01/2018	**Bachelor in Mathematics** Albert-Ludwigs-University of Freiburg, Germany Thesis: Formal Group Laws Minor: Management Final grade: 1.1 (scale: 1 best, 6 worst)
09/2016 - 02/2017	**Semester abroad at Universidad de Sevilla, Spain**

Research Interests

Causality, Graphical Models, High-dimensional Statistics, Algebraic Statistics.

Publications and Preprints

12 Leopold Mareis,**Nils Sturma**, Mathias Drton. Semiparametric Inference for Half-Trek Estimators in Linear Structural Equation Models. *Submitted*, `https://arxiv.org/abs/2606.26931`.

11 Tom Hochsprung, **Nils Sturma**, Jakob Runge, Mathias Drton, Andreas Gerhardus. Identifying Direct Causal Effects in Latent Factor Models by Accounting for Unidentified Parents. *Submitted*, `https://arxiv.org/abs/2605.28105`.

10 Benjamin Hollering, Pratik Misra, **Nils Sturma***. Efficient Symbolic Computations for Identifying Causal Effects. *Submitted*, `https://arxiv.org/abs/2604.20516`.

9 Shiva Madadkhani, **Nils Sturma**, Mathias Drton, Svetlana Ikonnikova. Mapping the Causal Structure of Price Formation in Texas's Transitioning Electricity Market. *Submitted*, `https://arxiv.org/abs/2604.14257`.

8 **Nils Sturma**, Mathias Drton. Trek-Based Parameter Identification for Linear Causal Models With Arbitrarily Structured Latent Variables. *Submitted*, `https://arxiv.org/abs/2507.18170`.

7 Dennis Leung, **Nils Sturma**. Singularity-Agnostic Incomplete U-statistics for Testing Polynomial Constraints in Gaussian Covariance Matrices. *Submitted*, `https://arxiv.org/abs/2401.02112`.

6 **Nils Sturma**, Miriam Kranzlmueller, Irem Portakal, Mathias Drton. Matching Criterion for Identifiability in Sparse Factor Analysis. *Psychometrika, 2026, Vol. 91, No. 2, 536-555.*

5 Mathias Drton, Alexandros Grosdos, Irem Portakal, **Nils Sturma***. Algebraic Sparse Factor Analysis. *SIAM Journal on Applied Algebra and Geometry, 2025, Vol. 9, No. 2, 279-309.*

4 Yulia Alexandr, Jane Coons, **Nils Sturma***. Mixtures of Discrete Decomposable Graphical Models. *Algebraic Statistics, 2024, Vol. 15, No. 2, 269-293.*

3 **Nils Sturma**, Mathias Drton, Dennis Leung. Testing Many Constraints in Possibly Irregular Models Using Incomplete U-Statistics. *Journal of the Royal Statistical Society Series B: Statistical Methodology, 2024, Vol. 86, No. 4, 987-1012.*

2 **Nils Sturma**, Chandler Squires, Mathias Drton, Caroline Uhler. Unpaired Multi-Domain Causal Representation Learning. *Advances in Neural Information Processing Systems 36, NeurIPS 2023, Spotlight.*

1 Rina Foygel Barber, Mathias Drton, **Nils Sturma***, Luca Weihs. Half-Trek Criterion for Identifiability of Latent Variable Models. *Annals of Statistics, 2022, Vol. 50, No. 6, 3174–3196.*

* Shared first authorship/ alphabetical order of the authors.

Awards/ Grants

Award for Good Teaching	Third place for best exercise session (High-dimensional Statistics, 2024). Awarded by the Mathematics Student Council of the TUM.
SIAM Student Travel Award	Competitive travel award for the 2023 SIAM Conference on Applied Algebraic Geometry.
MDSI/Linde PhD Fellowship	Competitive grant awarded to PhD students at TUM working on topics related to data science. The grant financed 50% of the 1.0 TV-L E13 position I held during my PhD studies.
Alumni-prize 2018	Every year the prize is awarded by Alumni Freiburg e.V. to the two most outstanding theses (Bachelor or Master) at the Faculty of Mathematics at the University of Freiburg.

Teaching Experience

Lecturer	Randomization and Causation at EPFL, course for students at the end of their Bachelor/ beginning of their Master, spring term 2026 (approx. 60 students).
Substitute Lecturer	In total four lectures of the master course Graphical Models at TUM (summer terms 2023 and 2025). One lecture of the master course High-dimensional Statistics at TUM (summer term 2024).
Teaching Assistant	Graphical Models at TUM, summer term 2025. High-dimensional Statistics at TUM, summer term 2024. Linear Algebra 2 at University of Freiburg, summer terms 2016 and 2017.
Master Seminars	Advances in Statistical Inference at TUM, winter term 2024/2025.
Mentorship	Orso Renucci, Bachelor project, 2026. Jannis Friebe, Master thesis, 2025. Daniels Birmans, Bachelor thesis, 2025. Mingyi Guo, Master thesis, 2025. Carolina Kornitzer, Bachelor thesis, 2025. Miriam Kranzlmüller, Interdisciplinary Project, 2024. Javier Yraola Meins, Bachelor thesis, 2023. Moritz Ebert, Master thesis, 2023. Julian Rittmaier, Bachelor thesis, 2022.
Upskilling	“ProLehre” Onboarding Course offered by TUM: Designed to prepare instructors for their teaching task. Elective Module: Supervising Students’ Theses.

Professional Activities

- Reviewer for Annals of Statistics, Biometrika, Electronic Journal of Statistics, Journal of Machine Learning Research, Bernoulli, Algebraic Statistics, La Matematica, IEEE Transactions on Signal Processing, NeurIPS conference, UAI conference, CLeaR conference.
- Part of the organizing committee for the 2025 *CLeaR conference.*
- Co-organizer of the Workshop *Causal Inference for Time Series Data* at UAI 2024.
- Co-organizer of the *European Workshop on Algebraic Statistics and Graphical Models* 2024.
- Co-organizer of the mini-symposium *Algebraic Methods in Graphical Models* at the SIAM Conference on Applied Algebraic Geometry 2023.

Software

- `SEMID`: An R-package that provides routines to check identifiability of causal effects in linear models. Developer and maintainer on `CRAN`.
- `TestGGM`: An R package for testing the goodness-of-fit of Gaussian graphical models. Lead developer, available on `GitHub`.

Talks & Presentations

04/2026	European Causal Inference Meeting, Oxford, UK. Poster presentation on *Trek-Based Parameter Identification for Linear Causal Models With Arbitrarily Structured Latent Variables.*
02/2026	Statistics Seminar at Binghamton University, USA. Invited talk on *Matching Criterion for Identifiability in Sparse Factor Analysis.*
12/2025	IMS International Conference on Statistics and Data Science, Seville, Spain. Contributed talk on *Matching Criterion for Identifiability in Sparse Factor Analysis.*
02/2025	Statistics Seminar at University College Dublin, Ireland. Invited talk on *Identifiability in Sparse Factor Analysis.*
08/2024	Bernoulli-IMS World Congress in Probability and Statistics, Bochum, Germany. Invited talk on *Identifiability in Sparse Factor Analysis.*
03/2024	Statistics Seminar at University of Melbourne, Australia. Invited talk on *Identifiability in Sparse Factor Analysis.*
12/2023	NeurIPS, New Orleans, USA. Poster presentation on *Unpaired Multi-Domain Causal Representation Learning.*

07/2023	SIAM Conference on Applied Algebraic Geometry, Eindhoven, Netherlands. Contributed talk on *Introduction to Algebraic Methods in Graphical Models.*
04/2023	Workshop on Causal Representation Learning, Tübingen, Germany. Contributed talk on *Unpaired Multi-Domain Causal Representation Learning.*
03/2023	YES Causal Inference Workshop, Eindhoven, Netherlands. Poster presentation on *Parameter Identifiability in Latent Variable Models.*
03/2023	German Probability and Statistics Days, Essen, Germany. Contributed talk on *Testing Many and Possibly Singular Polynomial Constraints.*
08/2022	17. Doktorand:innentreffen der Stochastik, Klagenfurt, Austria. Contributed talk on *Half-Trek Criterion for Identifiability of Latent Variable Models.*
06/2022	IMS Annual Meeting in Probability and Statistics, London, UK. Contributed talk on *Half-Trek Criterion for Identifiability of Latent Variable Models.*

Industry Experience

09/2018 - 02/2021	**Part-time internship in the Data Analytics department at Zeppelin** Development of machine learning applications for the construction industry. Focus on predictive maintenance, demand forecasting and process optimization.
04/2019 - 08/2019	**BMW & TUM Data Innovation Lab (project)** Deep Learning approach to predict lane changes using vehicle sensor data.
01/2017 - 03/2021	**Freelancer at Ernst-Klett-Verlag** Proofreading of mathematical school books.
02/2018 - 07/2018	**Internship at BMW** Controlling: inner-year targets of entire BMW Group.